READING PASSAGE 1

*You should spend about 20 minutes on **Questions 1–13**, which are based on Reading Passage 1 below.*

History of Film

Early Optical Toys and the Persistence of Vision

The illusion of moving pictures is rooted in two optical phenomena: persistence of vision and the phi phenomenon. Persistence of vision causes the human brain to briefly retain an image after it disappears, while the phi phenomenon creates an impression of continuous movement when images are shown in rapid succession.

Long before actual films, inventors built playful devices to exploit these effects. In the 1830s, devices like the phenakistoscope (1832), a spinning disc with sequential drawings viewed via slits and a mirror, and the zoetrope (1834), a rotating drum with slits cut into its sides, produced convincing illusions of motion from static images. The zoetrope's design—a cylinder with vertical slits and a strip of pictures on its inner wall—allowed multiple people to peer through the slits and see a "moving" scene as the drum spun. These Victorian-era optical toys demonstrated the principles that make cinema possible.

In 1877, French inventor Charles-Émile Reynaud introduced the praxinoscope, an improved successor to the zoetrope. Like the zoetrope, it used a strip of pictures inside a spinning cylinder, but the praxinoscope replaced the zoetrope's narrow viewing slits with an inner circle of mirrors. This clever modification meant a brighter, undistorted view of the images: as the wheel turned, a viewer looking into the mirrors saw a smooth animation without the dimming caused by slits. In essence, Reynaud's 1877 praxinoscope was "similar to a zoetrope, with mirrors in place of viewing slits," yielding a clearer picture of the moving drawings.

Such optical inventions—along with the earlier magic lantern (a 17th-century projector for painted slides) and thaumatropes (hand-twisted picture disks from the 1820s)—were the precursors of cinema, demonstrating that sequential images could create the illusion of motion.

Pioneers of Motion Photography (1870s–1880s)

Animation toys were limited to drawn images. The next step toward cinema was capturing real motion. This became possible with advances in photography. By the 1870s, photographic exposure times had dropped from an hour to a fraction of a second, and innovators began experimenting with series photography—taking rapid sequences of pictures to dissect motion.

One famous early experiment was conducted by English-born photographer Eadweard Muybridge. Beginning in 1872, Muybridge worked for Leland Stanford, a racehorse owner, to settle a debate: whether a galloping horse ever lifts all four hooves off the ground. Muybridge set up a row of cameras along a track, each triggered by the horse's hooves as it passed. In 1877 he succeeded—the resulting photographs showed the horse fully airborne mid-gallop. To demonstrate the result, Muybridge mounted the images on a rotating disk and projected them with a magic lantern, producing a "moving picture" of the horse at full gallop as it had actually occurred in life. This was a crucial proof-of-concept: real-life motion could be recorded and then played back as an animation.

Meanwhile, in France, physiologist Étienne-Jules Marey devised a way to capture movement with a single camera. In 1882 Marey built the chronophotographic gun, a camera shaped like a rifle that could take 12 pictures per second of a moving subject (originally, birds in flight). Marey's "gun" exposed images in rapid succession onto a rotating glass plate (later onto roll film). Although Marey's goal was scientific (analyzing motion too fast for the eye) and he only briefly attempted to project his images, his work further advanced the techniques needed for motion pictures.

By the late 1880s, another key ingredient fell into place: celluloid roll film. Inventor Hannibal Goodwin conceived the idea of flexible film, and industrialist George Eastman began mass-producing photographic celluloid film in 1889. Unlike glass plates or paper strips, the new celluloid film could hold thousands of images in a long, durable ribbon. This was crucial—it enabled the recording of extended sequences of photographs and thus the possibility of lengthy moving pictures, not just split-second motion studies.

The First Motion-Picture Cameras

With fast photography and roll film available, inventors around the world rushed to build devices to record and display moving pictures. An early breakthrough came from Louis Le Prince, a French-born inventor. Le Prince created a single-lens motion picture camera in the late 1880s and managed to shoot several short films in Leeds, England. Notably, in October 1888 he filmed *Roundhay Garden Scene*, a few seconds of moving imagery often

cited as one of the first true motion pictures. By 1889 Le Prince was using Eastman's new roll film in his camera. He planned a public exhibition of his films in New York City in 1890—a demonstration that could have inaugurated the age of cinema—but fate intervened: Le Prince mysteriously disappeared in 1890 while traveling in France, and his film exhibition never took place. As a result, Le Prince's contribution to cinema remained little known for decades.

In the end, credit for the first practical movie camera went to someone else. William Kennedy Laurie Dickson, a young Scottish engineer, was working in West Orange, New Jersey, at Thomas Edison's laboratories, and he built what came to be widely regarded as the world's first functional motion-picture camera. Edison, already famous for his phonograph (invented in 1877), had envisioned adding visual marvels to his audio device. In 1888 he tasked Dickson with developing a motion-picture camera.

Building upon the prior work of Muybridge and Marey, Dickson combined crucial technologies: a fast-stop camera shutter mechanism (adapted from a clock's escapement) to regulate the intermittent motion of the film, and perforated celluloid film to advance images in precise synchronization. By 1891, Dickson had succeeded in creating the Kinetograph camera, which could expose a rapid series of images onto the film strip. The Kinetograph's photographic system initially recorded about 50 feet of film running at roughly 40 frames per second—enough for a short sequence of action.

Edison's Kinetoscope: A One-Viewer Motion Picture

Thomas Edison's approach to the new medium differed from Le Prince's public exhibition idea. Edison had originally conceived of motion pictures as a visual adjunct to his phonograph, something individuals could privately watch as they would listen to a record. Accordingly, after Dickson developed the camera, Edison did not immediately pursue a projector for large audiences. Instead, Dickson also designed a personal viewing device called the Kinetoscope, essentially a wooden cabinet peep-show machine. Inside each Kinetoscope, a continuous loop of film ran over a light source and shutter, and by peering through an eyepiece on top, a single viewer could watch the movie play.

Edison's first Kinetoscope parlor opened in New York in April 1894, featuring banks of coin-operated machines—each showing a short film to one person at a time. The Kinetoscope quickly became a commercial success; by 1894–95, Kinetoscope parlors and arcade installations were spreading across the U.S. and even overseas. To supply content, Edison's team produced dozens of brief films (often simple scenes, vaudeville acts, or novelty performances) using a rudimentary studio called the Black Maria—a cramped, tar-paper shack on rails that could rotate to follow the sun for lighting.

However, the Kinetoscope's design had two significant drawbacks: the films were short, and, crucially, only one person could view the show at a time. Despite Edison's doubts about projection, other inventors saw that the real future of cinema lay in showing films to large audiences on a screen, rather than as solitary peepshows.

Notably, Edison also made a strategic error—he neglected to secure foreign patents for the Kinetoscope system. This allowed others in Europe to legally copy and modify his devices. By 1894, even as Edison's machines proliferated, innovators abroad were already experimenting with improved motion-picture apparatus. In London, for example, engineer William Friese-Greene and others had filed patents for projection devices and camera-projector hybrids around the same time. In Paris, a public demonstration of Edison's Kinetoscope in 1894 inspired two brothers—Auguste and Louis Lumière—to take the final leap into projected cinema.

The Lumière Brothers and the Birth of Cinema (1895)

It was the Lumière brothers of France who ultimately gave the world the first true cinema experience. Drawing on their background in photography (the Lumières were manufacturers of photographic plates), the brothers set out to develop a device that could record moving images and project them onto a screen for an audience. By early 1895, Louis and Auguste Lumière had invented their apparatus, which they named the Cinématographe.

At roughly 5 kilograms (11 lbs), this innovative machine was much smaller and lighter than Edison's cameras—Edison's electrically driven Kinetograph had weighed in at over 200–300 kg (reportedly more than 1,000 lbs) with its heavy motor and batteries. In contrast, the Lumière Cinématographe was hand-cranked and portable, making location filming feasible. Remarkably, the Cinématographe was a 3-in-1 device: it served as a camera for filming, could be used to print copies of the film, and, with a quick configuration change, became a projector to beam the motion pictures onto a screen. (In fact, the word *cinema* derives from this invention's name.)

On December 28, 1895, the Lumière brothers hosted a landmark event in Paris—the world's first commercial public film screening. Held at the Salon Indien du Grand Café in Paris, this demonstration is widely regarded as the birth of cinema as a commercial entertainment. For the astonished paying audience, the Lumières projected a series of short films shot with their Cinématographe, including the famous *Workers Leaving the Lumière Factory* and *Arrival of a Train at La Ciotat Station*. The Cinématographe's projected images of everyday life elicited gasps and applause; viewers marvelled at the realism of moving photographs coming to life on a screen. Unlike Edison's one-person peepholes, here an entire room could share the same moving picture experience—an immersive spectacle that would quickly spread around the globe.

The Lumières' invention proved not only technically sound but also practical: "Unlike the Kinetograph, which was battery-driven and weighed over 1,000 pounds, the Cinématographe was hand-cranked, lightweight (less than 20 pounds), and relatively portable." This portability had profound effects on the kinds of films made. Edison's heavy Kinetograph was mostly confined to a studio (hence his early films featured staged vaudeville acts or subjects that could perform in front of a stationary camera). The Lumière camera, by contrast, could be carried into the field. Early Lumière films were largely

documentary views (*actualités*) of real life: workers leaving a factory, trains arriving, people walking about in city streets. These short snippets had little narrative but captured the motion of ordinary life in real environments—a novelty for audiences of the time.

Within months of their first show, the Lumière brothers began dispatching teams of cameramen-projectionists around the world. These operators traveled to far-off places—from Moscow to New York, from India to Japan—filming local scenes and screening movies to enthralled audiences everywhere. As one historian notes, “because the Lumières sent their cameramen all over the world in search of exotic subjects, the Cinématographe became the founding instrument of distant cinemas in Russia, Australia, and Japan.” In 1896, just a year after their first show, the Lumières opened theaters in London, New York, Brussels, and other cities to exhibit their growing catalog of films.

Global Spread and Lasting Impact

The immediate success of the Lumière shows signaled that motion pictures were more than a passing curiosity. What began as scientific experimentation and parlour entertainment had evolved into a new artistic and commercial medium. By 1896–1897, projected films (whether on Lumière machines or rival projectors) became a sensation across Europe and North America, often featured in music halls and vaudeville programs. This period was sometimes called the “novelty period,” when the mere fact of moving images was the draw— theaters even advertised the machines (“Edison’s Vitascope” or “Lumière’s Cinématograph”) as the headline attraction.

But filmmakers quickly started to extend content beyond short glimpses. Within a few years, pioneers like Georges Méliès in France and Edwin S. Porter in America were introducing storytelling, special effects, and editing to films, turning them from minute-long scenes into longer narratives by the early 1900s.

In its infancy, cinema drew from existing traditions of photography, theater, and vaudeville, yet it rapidly developed a language of its own. By the early 20th century, the “animated photographs” of the 1890s had transformed into a thriving film industry. As more people paid to see movies, investors built studios and dedicated movie theaters, and techniques improved for lighting, camera movement, and film editing. What had started as a novelty of science became, in the new century, the world’s dominant form of mass entertainment. Originally a product of 19th-century scientific endeavor, cinema became a medium of mass entertainment and communication—one that would continue to evolve through sound, color, and beyond, but always rooted in those 19th-century “magic” inventions that first brought pictures to life.

Sources

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Wikipedia — *Praxinoscope*

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